

Impact Statement Card

Team Meru -- AI6 | AI Saturdays Lagos Cohort 10

Agricultural Extension RAG: Smart Retrieval for Farmers

Purpose

This project builds the retrieval component of a Retrieval-Augmented Generation (RAG) system designed to connect smallholder farmers in Sub-Saharan Africa with relevant agricultural extension guidance. Given a farmer's question, the system retrieves and ranks the 5 most relevant factsheets from a corpus of 695 documents spanning 13 crops, 7 institutional sources, and 21 countries. The intended downstream use is to feed these retrieved documents into a language model that generates a clear, actionable answer.

The system is not intended to replace human agricultural extension officers. It is designed to augment and accelerate the process of finding relevant information, particularly in contexts where extension officer-to-farmer ratios are prohibitively high.

Positive Impacts

Improved information access: Smallholder farmers currently face significant barriers in finding relevant agricultural guidance. A well-performing retrieval system can surface the right factsheet in seconds, reducing the time between question and answer from days (waiting for an extension visit) to minutes.

Better farming outcomes: When farmers receive accurate, relevant guidance on pest management, soil health, planting schedules, and post-harvest handling, they can make better decisions. Better decisions translate to reduced crop losses, improved yields, and greater food security.

Support for extension services: Agricultural extension officers can use the system as a reference tool, quickly finding authoritative documents to support their field recommendations. This is especially valuable for officers covering multiple crops or unfamiliar regions.

Knowledge equity: By making retrieval work across 13 crops and 21 countries, the system moves toward equitable access to agricultural knowledge regardless of which crop a farmer grows or which country they farm in.

Potential Risks

Retrieval errors: No retrieval system is perfect. When the system returns an irrelevant or misleading document at rank 1, a downstream answer generator may produce incorrect advice. In agriculture, wrong advice can lead to crop failure, wasted inputs, or harmful practices (e.g., incorrect pesticide application).

Uneven crop coverage: The system may perform better on well-represented crops (maize, rice, tomato) than on underrepresented ones (millet, cotton, cocoa). Farmers growing minority crops could receive lower-quality retrievals, reinforcing existing disparities in agricultural research attention.

Automation bias: Users may place undue trust in system outputs because they are generated by a machine learning pipeline. Extension officers who defer to the system rather than applying their own expertise could miss context-specific factors that the retrieval system cannot capture.

Language exclusion: The system operates in English only. Farmers who are not literate in English cannot use it directly, and any translation layer introduces additional error.

Temporal staleness: The corpus is static. As agricultural conditions change -- new pest invasions, new crop varieties, shifting climate patterns -- the system will continue to retrieve outdated documents unless the corpus is actively maintained.

Values

This project is guided by the principle that agricultural knowledge is a public good. Research-backed guidance produced by institutions like CGIAR and FAO exists to help farmers, and retrieval systems should make that guidance more accessible, not less. We prioritize:

- Accuracy over coverage: it is better to return 3 highly relevant documents than 5 marginally relevant ones.
- Generalization over memorization: the system must work on new questions about unseen topics.
- Transparency: every stage of the pipeline (bi-encoder, BM25 fusion, cross-encoder reranking) is independently evaluable.

Mitigation Strategies

Rigorous evaluation: We evaluate using nDCG@5 with graded relevance labels, which penalizes both missing relevant documents and ranking them in the wrong order. The zero-overlap train/test design ensures we measure true generalization.

Hard negative mining: During training, we deliberately surface difficult negative examples -- documents that are superficially similar to relevant ones but do not actually answer the query. This teaches the model to make fine-grained relevance distinctions rather than relying on surface-level topic matching.

Cross-topic ensemble: We train 5 cross-encoder models on different topic-based folds, ensuring no single model overfits to a subset of crops or regions. Averaging their scores produces more robust rankings.

Per-crop monitoring: In a deployment setting, retrieval quality should be monitored per crop and per region. If performance degrades for a specific crop, targeted data collection and model updates should be triggered.

Human-in-the-loop: The system should be deployed alongside, not instead of, human extension services. Extension officers should validate retrieved documents before advice reaches farmers, particularly for high-stakes decisions involving pesticide use, investment, or land management.

Limitations

- The system retrieves documents; it does not generate answers. The quality of any downstream answer depends on both retrieval quality and the answer generation model.
- Performance was evaluated on 200 test queries. While the zero-overlap design provides a strong generalization signal, 200 queries cannot cover the full diversity of farmer information needs.
- The system has no mechanism to express uncertainty. It always returns exactly 5 documents, even when no highly relevant document exists in the corpus.
- The factsheet corpus is fixed at 695 documents. Real-world deployment would require ongoing corpus maintenance and expansion.
- All evaluation was conducted offline. User-facing evaluation (do farmers find the retrieved documents helpful?) has not been performed.